Sensor Report

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Driverless

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***Abstract*—Autonomous Formula Student racecars require an integrated sensing and computing architecture capable of perceiving the surrounding environment, estimating the vehicle's position and orientation, monitoring vehicle dynamics, and controlling steering and braking actuators. The reliability of autonomous driving depends not only on the accuracy of individual sensors but also on their calibration, synchronization, integration, and fault-management mechanisms. This report presents a detailed technical analysis of the sensing and computing systems relevant to an autonomous Formula Student racecar. The hardware considered includes the Hesai Pandar40P three-dimensional Light Detection and Ranging (LiDAR) sensor, VectorNav VN-200 Global Navigation Satellite System/Inertial Navigation System (GNSS/INS), Stereolabs ZED 2i stereo camera, a separate monocular camera, and the NVIDIA Jetson AGX Orin computing platform. Supporting systems include inertial measurement units (IMUs), accelerometers, gyroscopes, magnetometers, wheel-speed sensors, Hall-effect sensors, optical encoders, steering-angle sensors, brake-pressure transducers, and actuator-feedback sensors. The report explains the physical operating principles and mathematical models of these devices, evaluates their applications and limitations in autonomous racing. Particular attention is given to coordinate-frame transformations, timestamp synchronization, sensor fusion, calibration, electrical and mechanical installation, fault diagnosis, and safe-state behaviour. Indicative procurement budgets and potential Indian-market sourcing channels are included for preliminary engineering planning. Exact prices, availability, and hardware specifications must be confirmed through current manufacturer documentation and supplier quotations before procurement.**

Keywords—Autonomous racing, Formula Student, LiDAR, GNSS, IMU, INS, stereo vision, monocular vision, sensor fusion, wheel-speed sensing, Hall-effect sensors, optical encoders, Jetson AGX Orin, steer-by-wire, brake-by-wire.

Introduction

An autonomous Formula Student racecar must perceive its surroundings, estimate its motion, identify the racing track, and execute steering, acceleration, and braking commands without continuous human control.

These operations depend on several sensing technologies. A LiDAR detects surrounding objects through laser reflections. Cameras capture visual information. An inertial measurement unit measures acceleration and angular velocity. GNSS receivers provide global positioning information. Wheel encoders measure wheel rotation, while steering and braking sensors provide feedback about vehicle control.

Understanding the components inside each sensor is essential because sensor performance, accuracy, calibration requirements, failure modes, and integration constraints arise directly from its construction.

Basic Components of a Sensor

Although different sensors use different physical principles, many share a common measurement architecture.

Generalized sensor measurement chain

Component	Function
Sensing element	Responds directly to a physical quantity such as light, force, acceleration, pressure, or magnetic field
Transducer	Converts the physical response into an electrical signal
Excitation circuit	Supplies electrical power or a reference signal when required
Signal conditioner	Amplifies, filters, or linearizes the measured signal
Analog-to-digital converter (ADC)	Converts analog voltage or current into numerical measurements
Microcontroller or processor	Applies calibration, compensation, and communication logic
Communication interface	Transfers measurements through CAN, SPI, I²C, UART, Ethernet, USB, or another protocol
Housing and connectors	Provide mechanical protection, mounting, electrical connections, and environmental sealing

Not every sensor contains all these elements as separate components. For example, an analog Hall-effect sensor may not contain an ADC, while a digital IMU often integrates the sensing elements, conditioning electronics, ADC, and communication controller into a single package.

Why Are Sensors Required in a Driverless Formula Student Racecar?

A sensor is a device that detects a physical quantity or environmental condition and converts it into a measurable signal. In a driverless Formula Student car, sensors provide the information required for the vehicle to perceive its surroundings, estimate its motion, monitor its mechanical systems and execute autonomous driving commands.

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A conventional car is controlled by a human driver who uses vision, hearing and physical feedback to understand the vehicle's state. A driverless car does not have a human driver to perform these functions. It therefore requires electronic sensors and computer-based algorithms to obtain and interpret the information necessary for safe and accurate operation.

Sensors act as the interface between the physical vehicle and the autonomous control system. Without reliable sensor measurements, the vehicle cannot determine where it is, identify track boundaries, estimate its speed or verify whether steering and braking commands have been executed.

How sensors support autonomous driving

The information from sensors is used throughout the autonomous driving pipeline as follows.

1. Environment sensing

LiDAR and cameras collect information about the track and surrounding objects.

2. Perception

Algorithms identify cones, obstacles and track boundaries.

3. Localization and state estimation

GNSS, IMU and wheel sensors estimate position, orientation and speed.

4. Planning and control

The system calculates the desired path, steering angle and speed.

5. Actuation and feedback

Steering and braking sensors verify the response and help detect errors.

These stages operate repeatedly while the vehicle is moving. The quality of the sensor data affects perception accuracy, localization, control stability and the ability to detect faults.

Importance of sensor accuracy and reliability

The mere presence of sensors does not guarantee safe autonomous driving. Sensor measurements can be affected by noise, bias, temperature, vibration, electromagnetic interference, environmental conditions and communication delays.

For example:

- **A dirty camera lens may prevent reliable cone detection.**
- **LiDAR reflections from some surfaces may produce missing or inaccurate points.**
- **An IMU bias may cause the estimated orientation to drift.**
- **A wheel speed sensor fault may produce an incorrect vehicle-speed estimate.**
- **A steering-angle sensor error may cause the controller to misinterpret the actual wheel position.**
- **A brake pressure sensor fault may prevent the controller from correctly assessing brake-system response.**

Consequently, sensors require suitable selection, installation, calibration, synchronization, diagnostics and validation. Where required by the system safety design, independent measurements and fault-detection mechanisms should be used to identify unreliable readings.

Classification of Sensors on Different Bases

Sensors can be classified according to several different criteria. No single classification is sufficient because the same sensor can belong to multiple categories simultaneously.

For example, a Hall-effect wheel speed sensor is an active electronic sensor in the common excitation-based classification, a non-contact sensor, a magnetic sensor, a digital-output sensor if it provides pulses, and a wheel-speed sensor according to its application.

The classifications below are useful for understanding sensor operating principles, selecting suitable devices and documenting a driverless racecar's instrumentation system.

Classification based on the energy source

Active sensors

In the traditional instrumentation classification, an active sensor generates an electrical output directly from the measured quantity without requiring a separate excitation supply for the sensing element.

Working principle: The physical phenomenon being measured generates an electrical voltage or charge. For example, a thermocouple generates a voltage due to a temperature difference, while a piezoelectric element generates charge when mechanically stressed.

Passive sensors

A passive sensor requires an external excitation source or electrical supply for its sensing mechanism or measurement circuit.

Working principle: The measured quantity changes a property such as resistance or capacitance. An external circuit measures that change and converts it into a useful signal.

Classification based on the physical quantity measured

This classification is particularly useful when developing the sensor list for a driverless vehicle.

Sensor category	Physical quantity	Examples	Application
Optical sensors	Light intensity or image information	Cameras, photodiodes, photoelectric sensors	Object detection and optical measurement

Distance and ranging sensors	Distance to an object	LiDAR, ultrasonic sensors	Obstacle detection and depth estimation
Inertial sensors	Acceleration and angular velocity	Accelerometers, gyroscopes, IMUs	Vehicle dynamics and navigation
Magnetic sensors	Magnetic field	Hall-effect, AMR, GMR and fluxgate sensors	Wheel speed, position and heading
Position and displacement sensors	Linear or angular position	Encoders, potentiometers, LVDTs	Steering and actuator feedback
Speed sensors	Rotational or linear speed	Wheel speed pickups, encoders	Vehicle speed and wheel-slip estimation
Pressure sensors	Fluid or gas pressure	Piezoresistive and capacitive pressure transducers	Brake-pressure monitoring
Force and strain sensors	Force, load or deformation	Load cells, strain gauges	Structural and actuator monitoring
Temperature sensors	Temperature	Thermistors, RTDs, thermocouples	Thermal protection and monitoring
Electrical sensors	Current, voltage or electrical power	Hall current sensors, shunts, voltage dividers	Electrical-system diagnostics
Navigation sensors	Position, velocity and time	GNSS receivers	Global localization and timing

Classification based on the output signal

Analog sensors

Analog sensors produce a continuously varying signal within their operating range.

Working principle: A change in the physical quantity produces a corresponding change in voltage, current, resistance or another continuous electrical quantity.

For example, an analog pressure transducer may produce an output voltage that increases with pressure.

Digital sensors

Digital sensors provide discrete digital values or states.

Working principle: The sensing element detects the physical quantity, and electronic circuitry converts the measurement into digital data or discrete switching states.

Classification based on the contact with the measured object

Contact sensors

Contact sensors require physical contact with the object or system being measured.

Non-contact sensors

Non-contact sensors measure without direct physical contact with the target.

Examples:

LiDAR, Cameras, Hall-effect proximity sensors

Classification based on the transduction principle

The transduction principle describes how the physical quantity is converted into a measurable signal.

Principle	Main physical mechanism	Examples
Resistive	Resistance changes with the measured quantity	Strain gauges, RTDs, thermistors
Capacitive	Capacitance changes due to geometry or dielectric properties	MEMS accelerometers, capacitive pressure sensors
Inductive	Magnetic flux or inductance changes	Variable-reluctance pickups, LVDTs
Electromagnetic induction	Changing magnetic flux induces voltage	Inductive speed sensors
Hall effect	Magnetic field generates transverse voltage	Hall-effect wheel speed sensors
Magnetoresistive	Magnetic field changes resistance	AMR, GMR and TMR sensors
Piezoelectric	Mechanical stress generates electrical charge	Vibration and dynamic-force sensors
Thermoelectric	Temperature difference generates voltage	Thermocouples
Optical	Light intensity, interruption, reflection or phase is measured	Cameras, photoelectric sensors, optical encoders

Radio-frequency / satellite timing	Received radio signals are processed to estimate timing and position	GNSS receivers
Time of flight	Signal travel time is used to calculate distance	Pulsed LiDAR

This classification is important because it explains not just what a sensor measures, but why its output changes when the physical quantity changes.

Classification based on the reference measurement

Absolute sensors

Absolute sensors provide a measurement referenced to a defined scale, origin or external reference.

Relative sensors

Relative sensors measure a change with respect to a previous state, local reference or another measurement.

Classification based on the measurement method

Deflection-type sensors

The measured quantity causes a physical or electrical displacement that is measured and related to the quantity.

Null-balance or force-balance sensors

A feedback mechanism applies a balancing force or signal to keep the sensing element near a reference position. The balancing effort is used to estimate the measured quantity.

Examples include force-balance accelerometers and certain precision measurement instruments.

Classification based on the number of sensing axes

Type	Description	Examples
Single-axis	Measures one direction or component	Single-axis accelerometer, one-axis rotary encoder
Two-axis	Measures two components	Two-axis accelerometer or inclinometer
Three-axis	Measures three spatial components	Three-axis accelerometer, gyroscope or magnetometer
Multi-sensor / multi-axis system	Combines multiple sensing elements and quantities	Six-axis IMU, nine-axis IMU, GNSS/INS

Classification based on the operating environment

General-purpose sensors

Designed for standard operating environments where exposure to extreme vibration, temperature, moisture or electrical interference may be limited.

Industrial or automotive-grade sensors

Designed and tested for more demanding environmental and electrical conditions, depending on the specific product rating.

Classification based on the application in a driverless racecar

Application group	Sensors included	Main purpose
Perception	LiDAR, monocular camera, stereo camera	Detect track geometry and obstacles
Localization	GNSS, IMU, INS, camera-based odometry	Estimate position and orientation
Vehicle-state estimation	Wheel speed sensors, accelerometers, gyroscopes	Estimate velocity and dynamic behaviour
Actuator feedback	Steering encoders, motor-position sensors, brake pressure sensors	Verify steering and braking response
Electrical monitoring	Current and voltage sensors	Monitor power and electrical health
Thermal monitoring	Thermistors, RTDs, thermocouples	Monitor component temperature
Structural monitoring	Strain gauges and load cells	Measure loading and mechanical response
Safety monitoring	Redundant position, speed, pressure and actuator feedback	Identify faults and support safe-state decisions

LiDAR Sensor

Example: Hesai Pandar40P



LiDAR stands for Light Detection and Ranging. It measures distances by transmitting laser light and detecting its reflection from surrounding surfaces.

Major components and their functions

Component	Construction and function
Laser emitters	Semiconductor laser diodes generate short optical pulses directed into the environment
Laser driver circuit	Supplies precisely controlled electrical pulses to the laser diodes
Transmit optics	Lenses and optical structures shape and direct the outgoing laser beams
Receive optics	Collect returning light and focus it onto the photodetectors
Photodetectors	Convert reflected optical energy into electrical signals
Analog front-end	Amplifies weak detector signals and suppresses electronic noise
Timing and ranging electronics	Determine the delay between laser transmission and return detection
Rotating scanning assembly	Changes the azimuth direction of the beams to provide surrounding coverage
Angular position encoder	Measures the scanner's rotational position so each distance measurement can be assigned a direction
Processing electronics	Calculate range measurements, organize point-cloud data, and handle communication
Ethernet interface	Transmits measurement packets to the vehicle computer

The Pandar40P is a multi-beam rotating LiDAR. Its exact internal optical arrangement, receiver implementation, and electronics are manufacturer-specific; the table describes the functional subsystems rather than an independently verified teardown.

Step-by-step working

1. **The laser driver energizes a laser emitter, producing a short light pulse.**
2. **Transmit optics direct the pulse toward an object, such as a track cone.**
3. **Part of the light reflects back toward the LiDAR.**
4. **Receive optics collect the returning light.**
5. **A photodetector converts the received light into an electrical signal.**
6. **The analog circuitry amplifies and conditions the signal.**

Timing electronics estimate the pulse's round-trip travel time.

The processor calculates the distance using:

\[R=\frac{c\Delta t}{2} \]

where \(R\) is the measured distance, \(c\) is the speed of light, and \(\Delta t\) is the round-trip time.

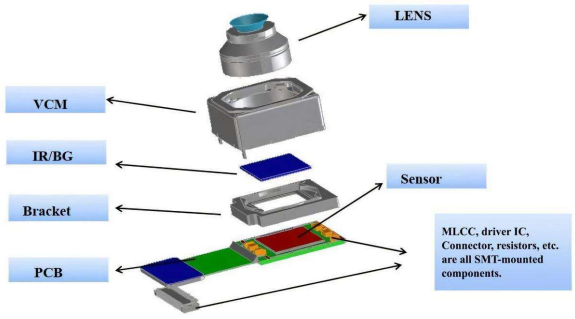
The angular encoder identifies the direction of the measurement.

The processor combines the measured distance and beam direction to generate a three-dimensional point.

Repeated measurements from multiple beams and successive scan angles produce a 3D point cloud. The LiDAR should be mounted on a rigid structure with an unobstructed field of view. Its coordinate frame must be calibrated relative to the vehicle frame

Camera Sensors

Monocular camera



A monocular camera uses one optical viewpoint to capture two-dimensional images.

Internal components

Component	Function

Lens assembly	Focuses incoming light onto the image sensor
Aperture	Controls the amount of light entering the camera
Optical filter	Suppresses unwanted wavelengths or protects image quality
CMOS image sensor	Converts incident light into electrical charge in an array of pixels
Pixel photodiodes	Accumulate charge in response to photons
Pixel readout circuitry	Reads the electrical signals generated by the pixels
ADC	Converts analog pixel signals into digital values
Image signal processor (ISP)	Performs operations such as demosaicing, noise reduction, and colour correction where supported
Exposure controller	Controls integration time and, where available, gain
Clock and synchronization circuitry	Coordinates frame capture and timing
Data interface	Transfers image data through USB, MIPI CSI, Ethernet, or another interface
Housing and mounting system	Protects and positions the camera

Working principle

Light reflected from an object passes through the lens and forms an image on the CMOS sensor.

Each pixel contains a photosensitive region. Photons generate charge carriers, and the accumulated charge is related to the incident light during the exposure period.

The readout circuit converts this charge into an electrical signal. The ADC produces digital pixel values, and image-processing circuitry prepares the frame for the vehicle computer.

Computer vision algorithms then identify cones, track boundaries, and other objects.

A monocular camera does not directly measure metric depth from a single ordinary image. Distance must be inferred using geometric assumptions, known object dimensions, motion across frames, or learned depth estimation.

Stereo camera



A stereo camera uses two viewpoints separated by a known baseline.

1. Components specific to stereo vision

Component	Function
Left camera module	Captures the scene from the left viewpoint
Right camera module	Captures the scene from the right viewpoint
Stereo baseline	Fixed physical separation between the two optical centres
Synchronization electronics	Coordinate image capture to reduce temporal mismatch
Calibration parameters	Describe lens distortion, intrinsic camera parameters, and relative camera geometry
Stereo processing software	Matches corresponding features between images and estimates disparity
Depth computation	Converts disparity into depth using camera geometry

2. Working principle

Both cameras capture the same scene from slightly different positions. Corresponding image features appear at different horizontal positions after stereo rectification.

The software calculates disparity \(d\).

Depth is estimated from:

\[Z=\frac{fB}{d} \]

where \(Z\) is depth, \(f\) is focal length in pixel units, \(B\) is the baseline, and \(d\) is disparity in pixels.

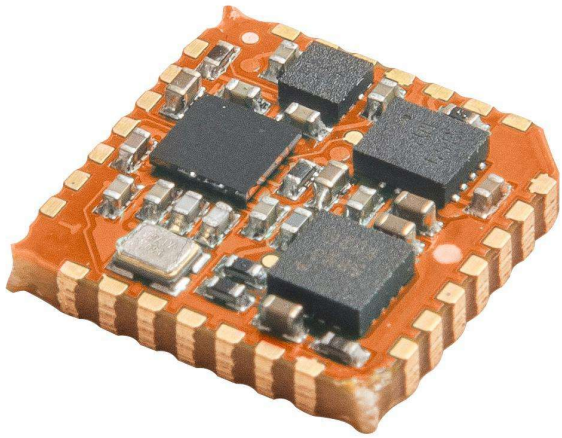
A larger disparity generally indicates a closer object.

ZED 2i example

The ZED 2i combines stereo image capture with onboard sensing capabilities, including inertial sensing. Its host-side software can provide depth, spatial perception, and visual-inertial functions. These capabilities depend on the specific SDK, configuration, and computing platform. For the racecar, the camera must be rigidly mounted, its lenses protected from contamination, and its calibration verified after any mechanical adjustment.

Inertial Measurement Unit (IMU)

An IMU measures motion-related quantities using accelerometers and gyroscopes. Some IMUs also include magnetometers.

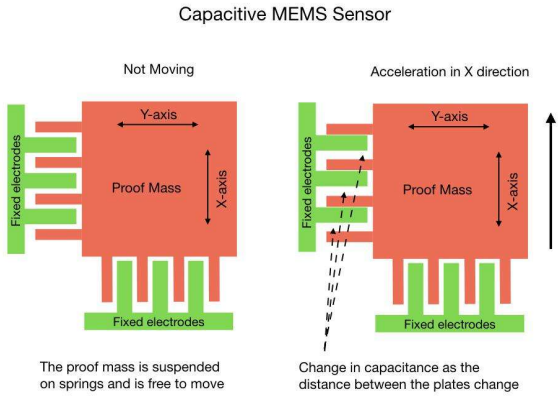


Components of an IMU

Component	Function
Three-axis accelerometer	Measures specific force along the X, Y, and Z axes
Three-axis gyroscope	Measures angular velocity about the X, Y, and Z axes
Magnetometer, if included	Measures magnetic field components
MEMS sensing structures	Mechanically respond to acceleration or rotation
Capacitive sensing electrodes	Detect tiny changes in capacitance caused by MEMS movement
Analog front-end	Amplifies and conditions sensor signals
ADC	Converts the signals into digital measurements
Temperature sensor	Monitors temperature for compensation
Calibration memory	Stores correction parameters for bias, scale factor, and axis alignment
Digital processor	Performs filtering, compensation, and sometimes orientation estimation
Clock and communication interface	Timestamp and transmit measurements

Accelerometer construction and working

A. Capacitive MEMS accelerometer



This is a widely used accelerometer type in automotive and robotics applications.

Main components:

Proof mass — a small movable silicon mass.

Suspension beams — flexible structures supporting the proof mass.

Fixed electrodes — stationary conductive plates.

Movable electrodes — electrodes attached to the proof mass.

Capacitive measurement circuit — detects changes in capacitance.

Signal-conditioning electronics — convert the capacitance change into acceleration.

Working principle:

When the sensor accelerates, inertia causes the proof mass to deflect relative to its frame. This changes the spacing or overlap between movable and fixed electrodes.

Capacitance is approximately:

\[C=\frac{\varepsilon A}{d} \]

where \(\varepsilon\) is permittivity, \(A\) is electrode overlap area, and \(d\) is separation. The electronic circuit measures the capacitance change and estimates acceleration.

The underlying mechanical relationship is: F=ma

A real accelerometer measures specific force, so gravity and vehicle orientation must be considered when interpreting its output.

Racecar application: Longitudinal acceleration during braking, lateral acceleration during cornering, and motion estimation.

Piezoresistive accelerometer

Components: Proof mass, suspension beams, piezoresistive elements, Wheatstone bridge, excitation circuit, amplifier, and housing.

Working: Acceleration deflects the proof mass, stressing the piezoresistive elements. Their electrical resistance changes. A Wheatstone bridge converts this resistance variation into a voltage.

Advantages: Suitable for measuring static and dynamic acceleration; useful in high-shock designs.

Limitations: Temperature sensitivity and the need for compensation.

Piezoelectric accelerometer

Components: Seismic mass, piezoelectric crystal or ceramic, preload structure, electrodes, charge amplifier, and housing.

Working: Dynamic acceleration causes the seismic mass to exert force on the piezoelectric material. The material generates electrical charge proportional to the applied dynamic force.

Advantages: Wide dynamic range and high-frequency vibration measurement.

Limitations: Conventional piezoelectric accelerometers cannot measure sustained static acceleration reliably.

Racecar application: Vibration analysis, structural monitoring, and powertrain diagnostics rather than primary static attitude estimation.

Force-balance accelerometer

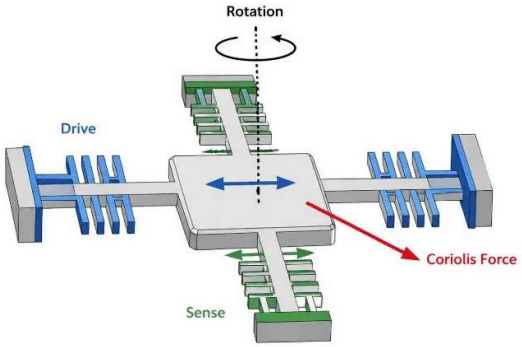
Components: Proof mass, position detector, feedback controller, electromagnetic or electrostatic actuator, and measurement electronics.

Working: The feedback controller applies a restoring force to keep the proof mass near its reference position. The required restoring force provides the acceleration measurement.

Gyroscope construction and working

A gyroscope measures angular velocity, generally expressed in degrees per second or radians per second.

3. A. MEMS vibratory gyroscope



Components:

- **Vibrating proof mass**
- **Drive electrodes**
- **Drive control circuit**
- **Suspension springs**
- **Sense electrodes**
- **Capacitive readout circuit**
- **Signal processor**
- **Temperature compensation circuitry**

Working principle:

The drive electrodes maintain oscillation of a small mass. When the sensor rotates, Coriolis acceleration acts perpendicular to the driven motion.

$$\mathbf{a}_c=2(\mathbf{v}\times\boldsymbol{\Omega})$$

where $\mathbf{v}$ is the vibrating mass velocity and $\boldsymbol{\Omega}$ is angular velocity.

The resulting secondary displacement is detected by the sense electrodes. The processing circuit estimates angular velocity.

Racecar application: Yaw-rate measurement during cornering, roll-rate estimation, and vehicle-state estimation.

Ring-laser gyroscope

Components: Closed optical cavity, mirrors, laser gain medium, photodetectors, and frequency measurement electronics.

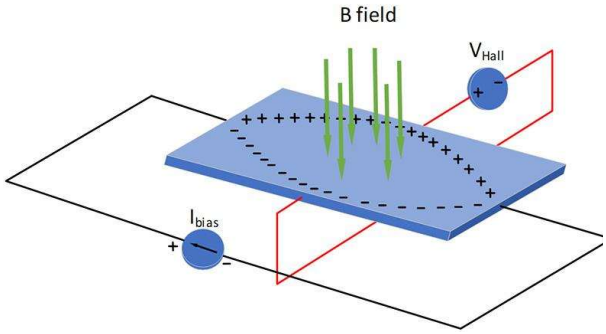
Working: Counter-propagating laser beams experience different optical path conditions during rotation, producing a measurable frequency difference related to angular velocity.

The IMU should be mounted on a rigid vehicle structure with a known orientation relative to the chassis. It should not be placed on a flexible panel that introduces unwanted motion.

Magnetometer Sensors

A magnetometer measures magnetic field strength or direction. Magnetometers can support heading estimation, although motors, power cables, ferromagnetic structures, and vehicle electronics can significantly distort the field.

Hall-effect magnetometer



4. Components

Component	Function
Semiconductor Hall plate	Carries electrical current and develops a transverse Hall voltage in a magnetic field
Bias current source	Supplies controlled current through the Hall plate
Hall voltage electrodes	Detect the voltage across the plate
Instrumentation amplifier	Amplifies the small Hall voltage
Offset-cancellation circuit	Reduces electrical offset
Temperature compensation circuit	Corrects temperature-dependent sensitivity
Comparator or ADC	Produces a switching signal or numerical magnetic-field measurement
Output driver	Transfers the signal to a controller

5. Working principle

When current flows through a semiconductor placed in a perpendicular magnetic field, moving charge carriers experience a Lorentz force. Charges accumulate on opposite sides of the semiconductor, producing a Hall voltage.

For an idealized Hall plate:

\[V_H=\frac{IB}{nqt} \]

where \(I\) is current, \(B\) is magnetic flux density, \(n\) is charge-carrier concentration, \(q\) is carrier charge, and \(t\) is plate thickness.

The voltage is conditioned and converted into either a digital switch output or an analog/digital field measurement.

Applications: Wheel speed sensing, magnetic position detection, motor commutation, and some steering position systems.

AMR magnetometer

AMR means anisotropic magnetoresistance.

Components: Ferromagnetic resistive elements, Wheatstone bridge, bias or set/reset structures where applicable, amplifier, ADC, and calibration electronics.

Working: The electrical resistance of the ferromagnetic element changes with the angle between its magnetization and the current direction. Bridge electronics detect the resistance change.

Advantages: Sensitive, compact, and useful for magnetic field sensing.

Limitations: Susceptible to magnetic interference and offset errors.

GMR and TMR magnetometers

GMR — Giant Magnetoresistance: Uses multiple thin magnetic and nonmagnetic layers. Electrical resistance changes with the relative magnetization orientation of magnetic layers.

TMR — Tunnel Magnetoresistance: Uses two magnetic layers separated by a thin insulating barrier. Electron tunneling resistance depends on the relative magnetization orientation.

Main components: Magnetic multilayer sensing element, bias circuitry, bridge or readout circuit, amplifier, ADC, and housing.

These technologies are particularly useful in compact magnetic position and angle sensors.

Fluxgate magnetometer

Components: Ferromagnetic core, excitation winding, sensing winding, oscillator, demodulator, and signal processor.

Working: An alternating excitation repeatedly drives the magnetic core into saturation. An external magnetic field changes the symmetry of the core's response, allowing the field to be estimated through harmonic detection.

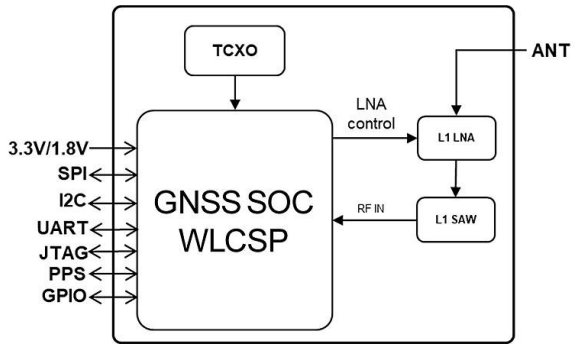
Advantages: High sensitivity, including to weak DC magnetic fields.

Limitations: Larger size, more complex electronics, and higher power consumption than many integrated magnetoresistive sensors.

GNSS Receiver

GNSS stands for Global Navigation Satellite System. It includes satellite navigation systems such as GPS, Galileo, GLONASS, and BeiDou.

A GNSS receiver estimates position and time from signals transmitted by navigation satellites.



Components

Component	Function
GNSS antenna	Receives weak satellite radio signals
Low-noise amplifier (LNA)	Amplifies received signals while minimizing added noise
RF filter	Suppresses unwanted radio frequencies
RF downconverter	Converts the received signal to an intermediate or baseband frequency
Local oscillator	Supplies reference frequencies for radio processing
ADC	Digitizes the received signal
Correlator and baseband processor	Detects satellite codes and tracks signal timing
Navigation processor	Calculates position, velocity, and time
Clock oscillator	Provides timing reference
Correction-data interface	Accepts differential or RTK corrections where supported
Communication interface	Sends navigation data to the vehicle computer

Working principle

1. **The antenna receives signals from multiple satellites.**
2. **The RF front-end filters and amplifies these signals.**
3. **The receiver identifies and tracks individual satellite signals.**
4. **Signal timing is used to estimate pseudoranges.**
5. **The navigation processor combines pseudoranges and satellite orbit information.**
6. **The receiver solves for three-dimensional position and receiver clock offset.**

A simplified pseudorange model is:

$$\rho_i=\|\mathbf{p}-\mathbf{s}_i\|+c\delta t+\epsilon_i$$

where $\mathbf{p}$ is receiver position, $\mathbf{s}_i$ is satellite position, δt is receiver clock error, and ϵ_i represents measurement errors.

RTK-capable GNSS systems can achieve substantially better positioning accuracy under suitable satellite visibility, correction availability, and operating conditions.

Inertial Navigation System (INS)

Example: VectorNav VN-200



An INS is a navigation system that estimates position, velocity, and orientation by integrating inertial measurements.

The VectorNav VN-200 is a GNSS-aided inertial navigation device. It combines inertial sensing, GNSS information, and onboard estimation algorithms.

6. **8.2 Components**

Component	Function
Three-axis accelerometer	Measures specific force
Three-axis gyroscope	Measures angular velocity
Three-axis magnetometer	Measures magnetic field
Barometric pressure sensor	Provides pressure information that may support altitude-related estimation
GNSS receiver	Provides satellite-based position, velocity, and time measurements
Microprocessor	Executes navigation and estimation algorithms

Calibration memory	Stores sensor calibration information
Timing circuitry	Coordinates measurements and timestamps
Communication interface	Outputs navigation data
Power conditioning	Provides stable internal operating voltages
Protective enclosure	Supports mounting and environmental protection

The exact hardware revision and supported functions should be confirmed against the relevant VN-200 manual.

7. 8.3 Step-by-step operation

1. The accelerometers measure specific force.
2. The gyroscopes measure angular velocity.
3. The navigation algorithm updates the estimated orientation.
4. The measured specific force is transformed into a navigation coordinate frame.
5. Gravity compensation is applied.
6. Acceleration is integrated to estimate velocity.
7. Velocity is integrated to estimate position.
8. GNSS measurements correct accumulated inertial navigation errors.
9. An estimation filter combines the measurements into a more stable navigation solution.

The basic relationships are:

$$\mathbf{v}(t)=\mathbf{v}(0)+\int_0^t\mathbf{a}(\tau)\,d\tau$$

$$\mathbf{p}(t)=\mathbf{p}(0)+\int_0^t\mathbf{v}(\tau)\,d\tau$$

These equations are simplified; a practical INS must account for orientation, gravity, Earth/reference frames, biases, and navigation dynamics.

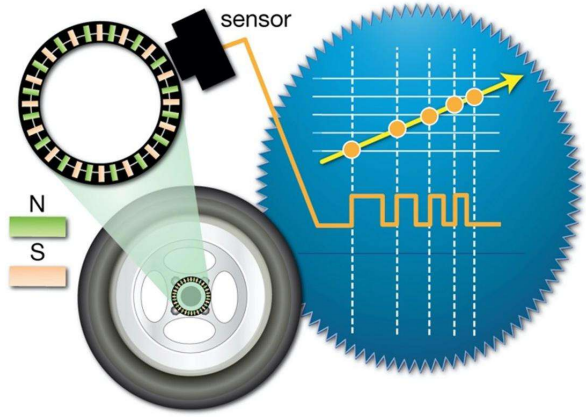
Advantages: Continuous motion estimation, rapid updates, and improved performance through GNSS aiding.

Limitations: Inertial drift during GNSS outages, magnetic interference, calibration requirements, and higher cost than a basic IMU.

Wheel Speed Sensors and Encoders

Wheel speed sensing is essential for estimating vehicle velocity, wheel slip, traction, and braking behaviour.

Hall-effect wheel speed sensor



Components

1. Permanent magnet or magnetic bias arrangement.
2. Hall-effect sensing element.
3. Ferromagnetic toothed wheel or multipole magnetic encoder ring.
4. Signal-conditioning electronics.
5. Comparator or digital output circuit.
6. Sensor housing, cable, and connector.

Working

As the wheel rotates, the teeth or magnetic poles pass the sensing element. This changes the magnetic field detected by the Hall element.

The signal-conditioning circuit converts the changes into pulses. The controller counts the pulses to calculate rotational speed.

Vehicle speed can be estimated as:

$$v=\frac{2\pi r(\text{RPM})}{60}$$

where r is the effective rolling radius.

Advantages: Can detect low rotational speeds, robust digital output, and relatively low cost.

Limitations: Requires correct alignment, air gap, and magnetic target compatibility.

8. 9.2 Variable-reluctance wheel speed sensor

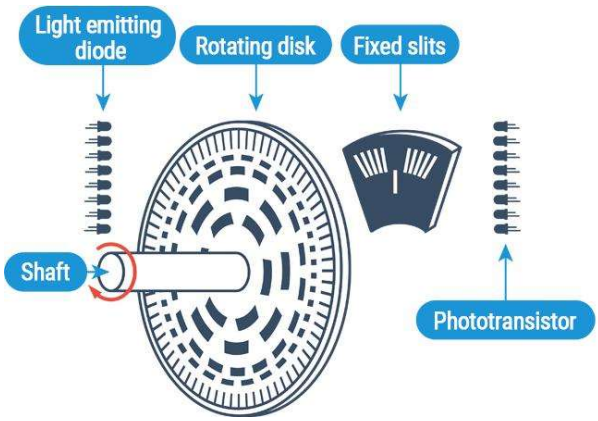
9. Components

- Permanent magnet
- Ferromagnetic core
- Copper sensing coil
- Rotating toothed target wheel
- Housing and electrical connections

10. Working

As teeth pass the magnetic core, magnetic flux through the coil changes.The generated AC signal has a frequency related to wheel speed.

Optical incremental encoder



11. Components

Component	Function
LED or light source	Produces a controlled optical beam
Slotted or patterned disk	Rotates with the shaft and modulates the light
Photodiode or phototransistor	Detects transmitted or reflected light
Comparator	Converts the detected light variations into digital pulses
Encoder electronics	Provides pulse outputs, often including A and B quadrature channels
Shaft and bearings	Support rotational movement
Housing	Protects the optical assembly

12. Working

As the disk rotates, alternating transparent and opaque regions interrupt the light beam. The photodetector generates a corresponding electrical signal.

In quadrature encoders, two channels are offset in phase, allowing the controller to determine rotation direction as well as count movement.

Advantages: High resolution and precise rotational measurement.

Limitations: Dirt, dust, vibration, and optical contamination can reduce reliability unless the encoder is adequately sealed.

Magnetic rotary encoder

Components: Rotating permanent magnet, Hall or magnetoresistive sensor array, angle-processing circuit, ADC, and digital interface.

Working: Rotation changes the orientation of the magnetic field relative to the sensing elements. The electronics estimate shaft angle from the measured field components.

Magnetic encoders are useful for steering-angle and actuator-position feedback because they can be compact and relatively resistant to optical contamination.

Absolute encoder

An absolute encoder provides a distinct position code for each shaft angle.

Components: Coded optical or magnetic track, sensing array, decoder, communication controller, and housing.

Unlike an incremental encoder, it can report absolute shaft position after startup without requiring a homing operation, provided its mechanical and electrical design supports that function.

13. X. Optical and Proximity Sensors

These sensors detect the presence, position, or movement of objects using light.

14. 10.1 Photoelectric through-beam sensor

Components: LED or infrared emitter, transmitter driver, receiver photodiode, amplifier, comparator, and housing.

Working: The transmitter directs light toward a separate receiver. An object interrupts the beam, changing the receiver output.

Application: Bench testing, actuator position verification, and test-rig automation.

15. 10.2 Retroreflective photoelectric sensor

Components: Light emitter, receiver, optical lens, reflector, signal-processing circuit, and output stage.

Working: Light travels from the sensor to a reflector and returns to the receiver. An object interrupts the return path..

16. 10.3 Diffuse reflective optical sensor

Components: Light source, transmit optics, receive optics, photodetector, amplifier, and comparator or distance-processing circuit.

Working: Light is projected toward an object, and reflected light is detected by the receiver.

The detected intensity depends on distance, surface reflectivity, and geometry. Simple diffuse sensors generally detect presence rather than accurately measuring distance.

17. 10.4 Optical time-of-flight proximity sensor

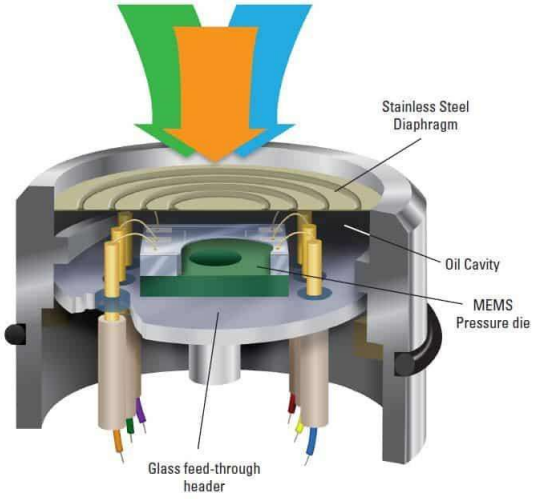
Components: Modulated or pulsed light emitter, optical receiver, photodetector, timing or phase-measurement circuit, processor, and optical window.

Working: The sensor estimates distance from the time delay or phase difference between transmitted and reflected light.

Application: Short-range obstacle or mechanism-position sensing, subject to the device's range and environmental limitations.

18. XI. Pressure Sensors

Pressure sensors are particularly relevant to hydraulic braking systems, pneumatic systems, and fluid monitoring.



19. 11.1 Piezoresistive pressure sensor

Working principle

1. **Brake fluid pressure acts on the diaphragm.**
2. **The diaphragm deforms elastically.**
3. **The deformation strains the piezoresistive elements.**
4. **Their resistance changes.**
5. **The Wheatstone bridge generates a voltage related to pressure.**
6. **Signal-conditioning electronics provide a usable output.**

Racecar application: Monitoring hydraulic brake pressure and checking whether commanded braking corresponds to measured pressure.

20. 11.2 Capacitive pressure sensor

Components: Flexible diaphragm, fixed electrode, reference cavity, capacitance-measurement circuit, and signal processor.

Working: Pressure deflects the diaphragm, changing the distance between the capacitor electrodes. The resulting capacitance variation is converted into pressure.

Advantages: High sensitivity and potentially low power consumption.

Limitations: Susceptibility to parasitic capacitance and the need for calibration.

Force, Load, and Strain Sensors

Strain gauge

Working

When a component deforms, the bonded strain gauge stretches or compresses. Its resistance changes because of changes in geometry and material properties.

The gauge factor is:

$$GF=\frac{\Delta R/R}{\varepsilon}$$

where ε is mechanical strain.

The Wheatstone bridge detects the small resistance variation.

Application: Suspension load measurement, structural testing, and validation of mechanical components.

Load cell

A load cell is a force transducer, commonly constructed using strain gauges bonded to an elastic mechanical element.

Working

Applied force deforms the elastic element. The strain gauges change resistance, and the bridge output is converted into force using calibration coefficients.

Racecar application: Brake-pedal force testing, actuator-force validation, and suspension measurement.

Temperature Sensors

Temperature monitoring is important for electronics, motors, battery systems, brakes, and sensor reliability.

Thermistor

Components

- **Temperature-sensitive semiconductor element**
- **Electrodes and leads**
- **Protective encapsulation**
- **Voltage-divider or resistance-measurement circuit**
- **ADC**

Working

The resistance of the semiconductor changes with temperature.

For an NTC thermistor, resistance decreases as temperature rises.The controller measures resistance and uses a calibration curve to estimate temperature.

Resistance temperature detector (RTD)

Components

- **Platinum or other metallic resistance element**

- **Insulating support**
- **Lead wires**
- **Protective sheath**
- **Precision current source**
- **Resistance-measurement electronics**

Working

The electrical resistance of the metal changes predictably with temperature. A Pt100, for example, has a nominal resistance of 100 Ω at 0°C.

Thermocouple

Components

- **Two dissimilar metal conductors**
- **Measuring junction**
- **Reference junction or terminal block**
- **Low-noise amplifier**
- **Cold-junction temperature sensor**
- **ADC**

21. Working

A temperature difference between the measuring junction and reference junction produces a thermoelectric voltage through the Seebeck effect.

Advantages: Wide temperature range and suitability for harsh environments.

Limitations: Small output voltage, noise sensitivity, and the need for cold-junction compensation.

Electrical Sensors

Hall-effect current sensor

Components: Magnetic core or flux concentrator where used, current conductor, Hall sensing element, amplifier, isolation structure, and output electronics.

Working: Current flowing through a conductor creates a magnetic field. The Hall element detects this field, and the electronics estimate current.

Advantages: Galvanic isolation is possible, and both DC and AC current can be measured.

Shunt-resistor current sensor

Components: Precision low-resistance shunt, Kelvin sense connections, differential amplifier, ADC, and protection circuitry.

Working: Current produces a small voltage drop across the resistor.

\[V=IR \]

The measurement circuit calculates current from the voltage drop and known resistance.

Advantages: Low cost and good accuracy with suitable design.

Limitations: Power dissipation, possible lack of isolation, and sensitivity to connection errors.

22. 14.2 Voltage sensor

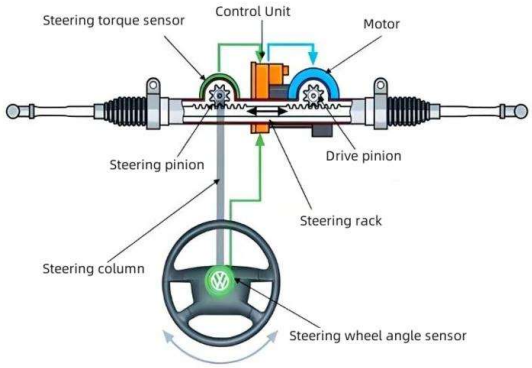
Components: Precision resistor divider, protection components, buffer or isolation amplifier, voltage reference, and ADC.

Working: A resistor divider reduces a higher voltage to a range compatible with the ADC. The processor reconstructs the original voltage using the divider ratio.

Application: Monitoring supply rails, low-voltage system health, and battery-related electrical parameters with appropriately rated hardware.

23. XV. Steering Sensors and Steer-by-Wire

Steer-by-wire is a control architecture in which electronic commands control a steering actuator. The sensing components provide feedback about the actual steering position and system condition.



24. 15.1 Main system components

Component	Function
Steering angle sensor	**Measures steering-wheel, column, rack, or road-wheel position depending on design**
Rotary encoder	**Measures actuator or shaft rotation**
Steering actuator motor	**Generates mechanical steering motion**
Motor driver	**Regulates current and power delivered to the actuator**

Steering mechanism	Converts motor movement into wheel steering motion
Electronic controller	Computes motor commands from the desired steering angle
Current sensor	Monitors actuator electrical load
Limit switch or end-stop sensing	Detects travel boundaries where installed
Emergency shutdown interface	Disables actuation when required by the safety system
Power and communication circuits	Supply energy and exchange commands and feedback

Steering angle sensor working

A magnetic steering angle sensor may use a permanent magnet attached to a rotating shaft and a Hall or magnetoresistive sensor. As the shaft rotates, the measured magnetic-field direction changes. The electronics calculate angular position.

Closed-loop steering control

1. The path-planning controller generates a desired steering angle.
2. The steering controller compares desired angle with measured angle.
3. The controller calculates an actuator command.
4. The motor driver energizes the steering motor.
5. The encoder measures the resulting motion.
6. The controller updates the command using feedback.

Safety requirements

Steering sensing should be checked for implausible values, loss of communication, stuck readings, excessive tracking error, and actuator overcurrent.

Where the applicable Formula Student rules require it, redundant sensing, independent monitoring, emergency shutdown, and defined safe-state behaviour must be incorporated.

Braking Sensors and Brake-by-Wire

Brake-by-wire or an autonomous braking system uses electronic control to command braking force while measuring the actual response.

Main components

Component	Function
Brake pressure sensor	Measures hydraulic pressure
Brake actuator	Applies force or displacement to the braking mechanism
Actuator motor or pneumatic drive	Provides mechanical energy
Position encoder	Measures actuator movement
Force sensor, if used	Measures applied mechanical force
Motor driver or valve driver	Controls the actuator
Electronic brake controller	Computes braking commands
Hydraulic master cylinder and lines	Transmit braking pressure in hydraulic systems
Brake calipers	Apply braking force to the discs
Safety monitoring circuit	Detects failures and initiates required protective actions

Step-by-step working

1. The autonomous controller requests braking.
2. The brake controller converts the request into an actuator command.
3. The actuator applies force or displacement to the braking mechanism.
4. Hydraulic pressure increases.
5. The pressure sensor measures the resulting pressure.
6. Wheel speed sensors provide information about wheel rotation and possible slip.
7. The controller checks whether measured braking behaviour is consistent with the command.
8. Safety logic detects pressure loss, excessive actuator travel, unexpected wheel behaviour, or communication failure.

NVIDIA Jetson AGX Orin — Processing Platform



The NVIDIA Jetson AGX Orin is not a sensor. It is an embedded computing platform that processes measurements received from sensors and runs autonomy algorithms.

Main components

Component	Function
CPU	Executes general software, control logic, and ROS 2 nodes
GPU	Accelerates parallel computations, including neural-network inference
Deep-learning accelerators	Support suitable AI inference workloads
RAM	Stores programs, images, point clouds, and intermediate data
Storage	Holds the operating system, models, configuration, and logs
Camera interfaces	Receive compatible camera data
USB and Ethernet interfaces	Connect external sensors and network devices
GPIO and other peripheral interfaces	Support compatible embedded connections
Power-management circuitry	Supplies regulated electrical power
Cooling assembly	Removes heat produced during operation
Carrier board	Exposes connectors, power input, and supported interfaces

Exact capabilities depend on the AGX Orin module, carrier board, software release, and hardware configuration.

25. **17.2 Working within the racecar**

1. **LiDAR sends point-cloud measurements.**
2. **Cameras send image frames.**
3. **IMU and INS devices send motion and navigation measurements.**
4. **Wheel speed and actuator sensors send feedback through suitable interfaces or controllers.**
5. **Perception software identifies cones and obstacles.**
6. **Localization software estimates the vehicle pose.**
7. **Planning software determines a path and desired motion.**
8. **Control software calculates steering and braking requests.**
9. **Dedicated vehicle controllers and safety circuits handle actuation and protective functions.**

The Jetson should not be treated as the sole safety-critical controller. Emergency-stop and other mandated safety functions require an architecture that satisfies the applicable rules and remains effective during a main-computer failure.

Comparison of Major Sensors

Sensor	Primary measurement	Important internal components	Main strength	Main limitation
LiDAR	3D distance and geometry	Laser, optics, detector, timing electronics, scanner	Direct spatial measurement	Cost and environmental sensitivity
Monocular camera	2D visual information	Lens, CMOS sensor, ADC, ISP	Rich colour and texture	No direct single-frame metric depth
Stereo camera	Images and estimated depth	Two cameras, synchronization, calibration, stereo processor	Visual depth perception	Disparity and lighting limitations
MEMS accelerometer	Specific force	Proof mass, springs, electrodes, readout	Compact and high update rate	Bias and vibration sensitivity
MEMS gyroscope	Angular velocity	Vibrating mass, drive and sense electrodes	Fast rotation measurement	Integration drift

Magnetometer	Magnetic field	Hall, AMR, GMR, or TMR element	Compact magnetic sensing	Magnetic interference
GNSS receiver	Position, velocity, time	Antenna, RF front-end, correlator, processor	Global reference	Satellite visibility and multipath
INS	Estimated navigation state	IMU, processor, aiding interfaces	Continuous navigation estimate	Drift without aiding
Hall wheel sensor	Wheel rotational speed	Hall element, magnet, target ring	Reliable pulse output	Air-gap and target dependence
Optical encoder	Shaft rotation	LED, patterned disk, photodetector	High resolution	Optical contamination
Pressure sensor	Fluid pressure	Diaphragm, sensing bridge, amplifier	Direct hydraulic feedback	Requires correct calibration and fluid compatibility
Load cell	Force	Elastic element, strain gauges, bridge	Quantitative force measurement	Installation sensitivity
Thermistor	Temperature	Resistive element, readout circuit	Inexpensive	Nonlinear characteristic

Procurement and Sourcing in India

The following sources are useful starting points for checking sensor availability, technical specifications, authorized distribution, and quotations.

Sensor or component	Manufacturer or sourcing website	Approximate price (INR)
Hesai Pandar40P LiDAR	Hesai	₹4,00,000–₹10,00,000+
Other Hesai LiDAR products	Hesai	₹1,50,000–₹15,00,000+
VectorNav VN-200 GNSS/INS	VectorNav	₹1,50,000–₹3,50,000
ZED 2i stereo camera	Stereolabs	₹50,000–₹90,000
NVIDIA Jetson AGX Orin developer kit	NVIDIA Jetson	₹1,50,000–₹2,50,000
MEMS sensors, encoders and electronics	Mouser India	₹200–₹25,000+ per component
Industrial sensors and development boards	element14 India	₹500–₹50,000+
Sensors, ICs and electronic components	DigiKey India	₹100–₹30,000+ per component
Educational and prototype modules	Robu.in	₹100–₹15,000+

For critical sensors, procurement should be based on exact model numbers, official datasheets, calibration documentation, connector specifications, operating-temperature limits, and manufacturer support.

Low-cost hobby modules may be appropriate for learning or bench experiments but should not automatically be considered suitable for safety-critical vehicle operation.

Conclusion

Sensors form the foundation of autonomous racecar perception, localization, motion estimation, and feedback control. However, understanding a sensor requires more than knowing its application.

Its internal components determine how it converts a physical phenomenon into a measurable electrical signal.

A LiDAR relies on laser emission, optical reception, and precise timing. A camera uses lenses, photodiodes, and image-processing circuits. A MEMS accelerometer uses a suspended proof mass and displacement sensing. A gyroscope detects rotational motion through Coriolis effects or optical principles. A GNSS receiver uses radio-frequency and timing electronics, while wheel encoders convert rotational movement into electrical pulses.

For a Formula Manipal driverless racecar, these technologies should be selected and integrated based on measurement requirements, accuracy, update rate, environmental robustness, computational demand, cost, and safety.

The most reliable architecture combines complementary sensors, validates their measurements, detects inconsistencies, and ensures that faults cannot result in uncontrolled vehicle actuation.

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